
A
FIXED INPUT
PRESPECIFIED
ASSAY-NATIVE READOUT
Fig. 2 + Fig. 3
prioritized genes
gene-level context
Program registry
prespecified display set
no reselection
Fixed before refit
25 proteins
descriptive set
Protein triage
open chromatin
B · D
ATAC · DIA-MS
spatial
E · F
Protein · histology
C
B
ρ = 0.46 · 66% direction-concordant
protein log2FC
mRNA log2FC
Not significant
Significant, concordant
Discordant
protein-significant (n = 211)
Discordant
23 (10.9%)
Concordant
188 (89.1%)
C
increasing severity
Fibrosis
NAS
adjusted NES
Steatosis
Ballooning
Inflammation
Fibrosis
NAS
Lipid droplet
+2.6
q=5e-6
PLIN2
ABHD5
PLIN4
PLIN1
ACLY
ECM / BM
+2.1
q=2e-6
P4HA2
AGRN
LAMA5
LAMB2
CD151
Immune / inflammation
+2.0
q=1e-2
CAPG
CASP1
CORO1A
CD74
HLA-DQA1
Detox
-2.1
q=4e-6
GGT1
MAT1A
ADH1B
CYP4A11
CYP2C19
AA catabolism
-2.2
q=9e-10
AASS
HGD
ACADSB
HAL
GNMT
Protein z-score
Partial Spearman rho
mRNA vs protein
mRNA log2FC
protein log2FC
padj < 0.05 and, for mRNA, |log2FC| > 0.5
otherwise
log2 FC
D
MASLD/liver fat (n = 22 genes)
Liver enzyme (n = 424 genes)
Hepatocyte
Stellate
Macrophage
Cholangiocyte
T/NK
Genes (%)
Both sources ≥50%
One source ≥50%
5% to <50%
Under 5% (incl. none)
E
Colocalized genes in open chromatin, n
direct
enzyme
Protein vs histologic burden partial ρ (95% CI)
robust
not robust
Hep
2 Sinusoidal endothelial
3 Oxidative phosphorylation
6 Hypoxia
7 Acute-phase / complement
11 Mitochondrial ETC
13 Nucleoside catabolism
14 Amino-acid metabolism
15 Coagulation factor
16 PPARa lipid metabolism
17 Hepatocyte stress
18 Amino-acid degradation
19 Fatty-acid / peroxisomal metab
21 Xenobiotic / drug metab
22 Ion / membrane potential
24 NRF2 antioxidant
25 UPR / ERN1 stress
29 Cell polarity
30 Immediate-early AP-1
Chol
2 Oxidative phosphorylation
3 Macrophage / immune
9 Junction / autophagy
14 Cell adhesion / actin
16 Mixed metabolic
19 Cytoskeleton / TGFB
25 FXR / bile-acid
Fib
1 Plasma protein
2 Oxidative phosphorylation
9 Hypoxia
10 Hepatocyte xenobiotic
12 S100 / mesenchymal
15 Hypoxia / VEGF
16 Actomyosin / MHC-I
17 Elastic-fiber ECM
19 ECM proteoglycan
23 Insulin response
27 TGFB / collagen
not scored
Mac
1 Myeloid housekeeping
5 Acute-phase / MT
6 Vesicle / signaling
12 Resident macrophage
14 Adhesion / Rho
15 Endocytosis / TLR2
T
1 Plasma protein
4 Translation / ribosome
8 Memory / MAIT
not scored
F
Stromal ECM
Adjusted program score (z)
Stromal ECM
Ductular injury
Guilliams et al.
Vu et al.
q=.001
q<.001
q=.079
q=.316
Moran excess over matched genes